

Strategic Carbon Response: What Q and P Establish Together

Q A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability [arXiv:2608.29130](#)

P Emission-Forecasting-Based Spatial-Temporal Carbon Response: A Multi-Agent Attention-Enhanced Deep Learning Framework [arXiv:2607.26560](#)

Research phase: PAUSE after 1 round, 0 ITERATE, 0 REVISE

The pair

Q designs quadratic payments that implement a separable, strongly concave resource-allocation problem with aggregate capacity limits at a Nash equilibrium, and proposes stochastic learning for that game. P forecasts day-ahead nodal carbon intensity and uses it to schedule mobile energy storage systems (MESSs) and distributed data centers (DDCs) on a modified IEEE 33-bus network. [\[ledger 1, 2\]](#)

The connexion pursued

The initial idea was to use Q's mechanism to make self-interested participants carry out P's carbon-aware dispatch. This is a research question rather than a direct application: P's agents are forecasting-workflow operators, not strategic MESS or DDC owners, and its dispatch model differs from Q's allocation problem. [\[ledger 1, 2, 26\]](#)

What was found

Q assumes convex local strategy sets, separable private valuations, continuous requests, and componentwise aggregate upper-capacity constraints. P minimizes forecast-weighted emissions subject to a cross-DDC workload-balance equality and binary MESS location and travel decisions. Q's implementation theorem therefore does not cover P's full dispatch problem. Recasting participants as owners of splittable workload batches could produce a convex restricted problem, but it would define a new market; a mandatory batch also excludes the zero action used in Q's individual-rationality proof. [\[ledger 1, 2, 12, 14\]](#)

For a restricted problem with fixed nonnegative forecast $\hat{e}$, nonnegative carbon weight β , and $N > 1$, the following charge has the algebraic properties needed to add a forecast-carbon signal:

$$\tau_n^C(x) = \beta \hat{e}^\top x_n - \frac{\beta}{N-1} \sum_{m \neq n} \hat{e}^\top x_m.$$

Its sum is zero at every action profile, while agent n 's own marginal charge is $\beta\widehat{e}$. Combined with Q's payment, it gives the best responses for effective valuations $u_n(x_n) - \beta\widehat{e}^\top x_n$, conditional on Q's other assumptions. It implements a *welfare and forecast-carbon tradeoff*, not P's pure carbon-minimization objective. Individual rationality transfers only where a feasible zero or outside action and Q's other individual-rationality premises hold. [ledger 8, 10, 11, 14, 16, 17]

The central carbon question remains counterfactual. P's carbon-flow equation makes nodal intensity depend on dispatch, whereas its scheduling objective uses forecast intensity as fixed coefficients. For realized emissions $C(x) = e(x)^\top x$, the marginal signal is

$$\nabla C(x) = e(x) + J_e(x)^\top x.$$

A fixed-intensity charge omits the second term. This observation diagnoses a possible gap between implementing a forecast objective and reducing realized emissions; it does not establish that P's reported simulation reduction is wrong. P describes a closed-loop assessment and reports a 33.86% reduction between its Cases 2 and 3, but the supplied text does not settle how robust that result is to strategic private costs and counterfactual forecast error. [ledger 3, 5, 17, 24, 26]

A narrow certificate is available for any implemented dispatch x^M and comparator x^0 . If each decision's forecast-carbon error is at most δ , then

$$C(x^0) - C(x^M) \geq \widehat{C}(x^0) - \widehat{C}(x^M) - 2\delta.$$

Thus a forecasted reduction exceeding 2δ certifies a realized reduction. Historical forecast MAPE alone does not supply the required error bound under alternative dispatches. Nor does welfare optimality alone preserve P's reported emissions percentage. [ledger 13, 15, 17]

Objections and their status

Two source-specific issues prevent treating Q's stochastic-learning result as ready for transfer. The reduced payment printed beside Q's learning update has an own-price quadratic coefficient with the opposite sign from its full payment; direct expansion gives the corrected blocks recorded in the ledger. Also, Q's theorem requires growing sample sizes, while its numerical schedule $Q_i = \lceil 0.96^{i+1} \rceil$ equals one throughout. These are textual inconsistencies requiring correction and verification, not proofs that the simulations fail. [ledger 6, 7, 9]

For dispatch-dependent, nonseparable carbon, the simple off-equilibrium balanced rebate cannot generally give every participant the full realized-carbon marginal. The ledger's two-agent example $C(x) = x_1 x_2$ yields a contradiction by mixed differentiation. This limits that rebate construction, not all possible mechanisms: Q requires budget balance only at equilibrium. [ledger 11]

Outside sources and novelty

The prior-work check identified incentive-compatible spatial data-center demand response by Chen and colleagues (<https://ieeexplore.ieee.org/document/9330574/>), equilibrium analysis of carbon-aware load shifting and average-intensity signals by Jiang and colleagues (<https://arxiv.org/html/2504.07248v2>), and joint electricity-carbon pricing with budget balance by Chen and Zhao (<https://arxiv.org/abs/2308.08195>). The ledger also records work on marginal emission signals and uncertain carbon-aware scheduling. Consequently, neither a generic incentive layer nor the average-intensity feedback critique is established here as novel. The material supports a bounded research program, not yet a publication claim. [ledger 4, 18, 19, 20, 22, 24]

What remains open and the smallest next step

A joint result still needs a valid strategic model for P's cross-DDC equality and MESS logistics, an implementation and participation argument for that model, and counterfactual emission evidence under dispatch-conditioned carbon-flow calculations. The smallest decisive next step is to fix a convex DDC-only subproblem, specify who owns each decision and its outside option, correct and verify Q's payment and learning update, then recompute realized carbon for the strategic dispatch and a comparator on the same IEEE 33-bus instance. A measured counterfactual error bound would show whether the predicted reduction survives strategic implementation; failure at this restricted step would rule out the proposed direct transfer.

[[ledger 20, 24, 26](#)]